

Fault-Tolerant Dynamic Scheduling and Routing for TSN based In-vehicle Networks

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Abstract—Functional safety is considered as one of the utmost requirements of the future autonomous vehicle, which consumes an ample amount of safety-critical data exchange among sensors, actuators and controllers through the in-vehicle network (IVN). IVN should provide enough guarantees to satisfy the high-reliability requirement of such autonomous vehicle applications. To avoid packet failures, Time-sensitive Network (TSN) proposes a fault-tolerant mechanism called frame replication and elimination for reliability (FRER) in IEEE 802.1CB standard. The main idea of FRER is to transmit safety-critical data via multiple disjoint paths between source and destination so that if one path fails to transmit the safety-critical data, the other path can still deliver the packet to its destination. In this paper, four different dynamic scheduling and routing heuristics are analyzed to support FRER functionality in TSN based IVN. These heuristics use combined score value, each score value is dedicated for a single path, as a measure to schedule and route incoming flow in multiple redundant paths on the fly. One of the algorithms, the Fault-Tolerant Bottleneck Heuristic (FTBH) outperforms others in terms of schedulability and response time. It schedules around 3.0 – 7.0% more flows as compared to other developed heuristics depending on the network load.

Index Terms—time-sensitive network (TSN), frame replication and elimination (FRER), in-vehicle network (IVN), vector bin packing (VBP), area control unit (ACU), telematic control unit (TCU), gate control list (GCL)

I. INTRODUCTION

As the automotive industry cruise towards the development of the fully autonomous vehicle, the development of such vehicles brings new challenges for the vehicle controllers, vehicle sensors as well as in-vehicle network design. Compared to today's vehicles, autonomous vehicles require new types of advanced features and functionalities that require more powerful processing at the vehicle computer as well as more reliable and high bandwidth in-vehicle communication networks. Some of these advanced features are only required to be active in a certain driving situation. For example, adaptive cruise control is activated when traffic is congested, some vehicle-to-X (V2X) applications [1] (e.g., platooning, trajectory exchange, etc.) can be activated depending on the driving conditions applicable to vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I) and vehicle-to-network (V2N) scenarios, truck trailer systems, and over the air (OTA) software updates. To meet the requirements of such new evolving self-adaptive features, the in-vehicle network has to be configured dynamically on the fly. These

new applications come up with a huge amount of safety-critical data which is required to deliver to the end system within a certain required deadline and reliability. Failures in the in-vehicle network can occur at any time, which could lead to the loss of safety-critical data. Therefore, the network must be dependable to guarantee the successful delivery of safety-critical data. To support fault tolerance, higher bandwidth, and flexibility in an in-vehicle network (IVN), Ethernet with Time Sensitive Network (TSN) extension is considered as a next-generation communication standard.

In our previous work [2], single-path dynamic scheduling and routing algorithms were designed to schedule the incoming flows on the fly efficiently. This work is an extension of the previous work, in which the algorithms provide fault-tolerant dynamic schedules and routes at runtime considering the zonal architecture of the IVN. The algorithms are part of a centralized network controller (CNC), which re-configures the gate-control list (GCL) of every egress port in the TSN switches using the network configuration protocol (NETCONF).

The remainder of the paper is organized as follows. Section II provides the introduction of TSN standards that are used in this paper. Section III presents a survey of the related work. The system model is introduced in Section IV. Section V presents problem formulation and four different heuristics for scheduling and routing with fault-tolerant extension. Section VI analyzes experimental evaluations. Finally, the paper is concluded in section VII.

II. TIME-SENSITIVE NETWORKS (TSN)

Time-Sensitive Networking (TSN) is a set of IEEE standards that extend the capabilities of traditional Ethernet to support bounded end-to-end delay, low packet jitter, and low packet loss for in-vehicle applications. The highest level of determinism for critical traffic is achieved with the help of standard i.e., IEEE 802.1 Qbv [3]. The IEEE 802.1 Qbv introduces time-aware shaper (TAS) capability and works together with IEEE 802.1 ASRev [4], which is a time synchronization protocol among all network devices. The TAS is deployed to every egress port of the network device and it is composed of eight priority queues. The incoming flows are placed in a transmit queue respective to their priorities. The time-aware gates (controlled via GCL) are connected to every

queue and have opening and closing functionalities. The flow is eligible for transmission when the gate of that queue is opened. The global schedule, implemented in a centralized network controller (CNC), is deployed to port specific gate control list (GCL), which sends a control signal (open and close) to time-aware gates.

The TSN task group developed IEEE 802.1CB [5] (Frame Replication and Elimination for Reliability (FRER)) standard for zero packet loss transmission, particularly for safety-critical flows. The source node or switching device generates and encode sequence number for every outgoing packets of the flow and replicate the packets into two or more depending on the redundancy level of the flow. It transmits the replicas through disjoint paths towards the destination. At the destination node, the receiving packets are examined with the sequence number and eliminate the packets if they are received previously from another disjoint path. In case, if one of the disjoint path fails to transmit the packets, the other disjoint path can still deliver the packets to its destination.

III. RELATED WORK

In recent years, some of the seminal work has been done in designing the static schedules and routes jointly [6] [7]. Zheng et. al. [8] presents the algorithm of dynamic insertion and removal of the tasks from time-triggered (TT) schedule in the avionics network. The author in [9], presents for the first time the reconfiguration of GCL at runtime such that delay constraints are satisfied. The author in [10], presents virtual machine migration as a use-case and proposed offline and online scheduling and routing algorithm in TSN's based time-aware shaper network. The offline schedule is designed based on a minimum distance tree (MDT), which considers only the potential dynamic targets. Nasrallah et. al. [11] has proposed two adaptive algorithms for time-aware shaper i.e., adaptive bandwidth sharing (ABS) and an adaptive slotted window (ASW). In the ABS, transmission for other traffic classes are allowed only if the scheduled queue is empty. The main idea of ABS is to further reduce the delay of scheduled traffic while keeping the delay of other traffic classes relative low. In ASW, the algorithm adaptively shifts the gating ratio to guarantee the burst of scheduled traffic. The author in [12], designed dynamic reconfiguration algorithms for time-aware switch within the context of IEEE 802.1Qcc management protocol. Their designed reconfiguration algorithms increase the admission of scheduled traffic at the delay cost of low priority traffics. In [13], the author evaluates the reliability of TT communication based on FRER by injecting the faults. The author in [14], exploited temporal redundancy and proposed the reliability-aware routing for AVB traffic. The problem is formulated as an ILP with the objective of attaining the maximum transmission reliability under transient errors. In this work, we have designed four fault-tolerant based dynamic scheduling and routing heuristics for TSN based IVN.

IV. SYSTEM MODEL

In this work, the network topology is modelled as unweighted directed graph $G = \langle \alpha, \beta \rangle$, where α is the set of network nodes, having switching capabilities, each node deems as the set of ACUs, CPUs, and TCU (Telematics control unit) and β is the set of links connecting ACUs, CPUs, and TCU. For example, a link $[\alpha_{acu1}, \alpha_{acu2}]$ of β transmit flows from source node ACU1 to the destination node ACU2 and similar a link $[\alpha_{acu2}, \alpha_{acu1}]$ of β transmit flows in opposite direction. The set of inserted flows is denoted by F_a and characterized by a tuple $\langle src_j, dst_j, T_j, w_j, r_j \rangle$, where src_j is the source node, dst_j is the destination node, T_j is the periodicity, w_j is the size of the flow and r_j is the redundancy level of the flow. The redundancy level determines how many copies of the flows are transmitted in the disjoint paths. T_{min} and T_{max} are the minimum and maximum period that the network support. The set of egress ports of switching devices and end systems is denoted by E e.g. $E = e_1, e_2, \dots, e_l$, wherein an egress port and respective ingress port constitute a link.

Similar to work [2], the flow placement is modelled as a vector bin packing (VBP), in which the items are viewed as flows with d -dimensional resource demand vector. The resource demand vector tells the required egress ports of the network. There may be multiple resource demand vectors of a flow and it depends on the network topology. Multiple resource demand vectors are useful in case the flow transmission requires multiple disjoint redundant paths. The transmission time of the flow is considered as the size of the flow. The minimum-period T_{min} partitions the maximum-period T_{max} into bins $[i.T_{min}, (i+1).T_{min})$. There are T_{max}/T_{min} number of bins that are static and known at design time. Taking Fig. 1a into account, it can be observed that there are twenty-four egress ports in the network. These egress ports in the network are modelled as the dimensions of the bin, i.e., dimensions from 1 to 24, as shown in Fig. 1b.

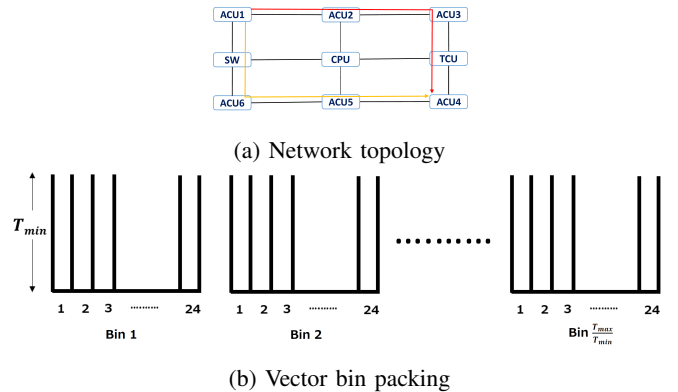


Fig. 1: Network topology and flow placement model

V. PROBLEM FORMULATION

The new network configuration is calculated on top of the existing network configuration without jeopardizing the

existing one. Each time the CNC receives the request of inserted $\mathcal{F}_a$ of certain redundancy level and de-inserted $\mathcal{F}_d$ flows from the existing running configuration, which is denoted by $\mathcal{S} \equiv (\mathcal{F}, \mathcal{D}, \mathcal{B}_s, \mathcal{R}_j)$, where $\mathcal{F}$ represents set of flows that configured in the network, $\mathcal{D}$ maps the flow to the egress ports of the network, $\mathcal{B}_s$ is the current state of the bins and $\mathcal{R}_j$ is the redundancy level of the configured flows in the network. Based on the requested flows, the algorithms determine new configuration $\mathcal{S}' \equiv (\mathcal{F}', \mathcal{D}', \mathcal{B}'_s, \mathcal{R}'_j)$.

A. Dynamic Scheduling and Routing Heuristics

In this section, we present four heuristics that are designed to solve the single-path [2] and their extension to multi-path (i.e. fault-tolerant) packet scheduling and routing problem in IVN. The heuristics evaluate a score value, which measures how well the flow j with path k fits to the bin i . The score value is evaluated only when all of the constraints are met (i.e., Capacity, Instance, Path, Periodic and Delay constraints [2]). The list of variables that are used in score value evaluation is shown in Table I.

The score value of the modified dot product heuristic is calculated by taking into account the resource demand of flow j and the remaining capacities of dimensions of bins. Mathematically, it is formulated as follows:

$$Score_j^{km} = \sum_{i=1}^I \frac{\sum_{l=1}^L \mathcal{D}_j^{kl} \times V_i^l(t) \times w_j}{\sum_{l=1}^L \mathcal{D}_j^{kl}} \times \mathcal{B}_{ij}^m, \quad \forall j \in J, \forall k \in K, \forall m \in M \quad (1)$$

The score value of modified most loaded heuristic is formalized as follows:

$$Score_j^{km} = - \sum_{i=1}^I \sum_{l=1}^L \frac{\mathcal{D}_j^{kl} \times w_j}{V_i^l(t)} \times \frac{\mathcal{B}_{ij}^m}{\sum_{l=1}^L \mathcal{D}_j^{kl}}, \quad \forall j \in J, \forall k \in K, \forall m \in M \quad (2)$$

The score value of the bottleneck heuristic is evaluated as the minimum remaining capacity of the required dimensions of all required bins. This heuristic also takes into consideration the flow path length. Mathematically, this heuristic is formalized as follows:

$$Score_j^{km} = \min_{\forall i \in I, \forall l \in L} S_i^l(j, k, m), \quad \forall j \in J, \forall k \in K, \forall m \in M \quad (3)$$

where,

$$S_i^l(j, k, m) = \begin{cases} \frac{\mathcal{D}_j^{kl} \times V_i^l(t) \times \mathcal{B}_{ij}^m}{\sum_{l=1}^L \mathcal{D}_j^{kl}}, & \text{if } \mathcal{D}_j^{kl} + \mathcal{B}_{ij}^m = 2, \\ \infty, & \text{otherwise} \end{cases} \quad (4)$$

The score value of the coefficient of variation heuristic is evaluated from standard deviation σ_j^{km} and mean load μ_j^{km} of the bins. Mathematically, it can be expressed as follows:

$$Score_j^{km} = - \frac{\sigma_j^{km}}{\mu_j^{km}}, \quad \forall j \in J, \forall k \in K, \forall m \in M \quad (5)$$

1) *Fault-tolerant Extension*: To provide reliability, multiple disjoint paths are used in FRER for the transmission of a single flow j . As there are multiple disjoint path pairs for a single flow j , the best disjoint path pair is selected based on the combined score value. The combined score is calculated by adding the score values of two or more disjoint paths of a flow j . The combined disjoint path set and bin set which has the maximum combined score value is selected for schedule and route. Mathematically, it is expressed as follows for redundancy level of one:

$$C_Score_j^m(k, k') = Score_j^{km} + Score_j^{k'm}, \quad \forall j \in J, \forall k \in K, \quad \forall m \in M, \forall k' \in K \quad (6)$$

where, k' and k are the disjoint paths of a flow j .

TABLE I: List of Variables

Symbol	Definition
$\mathcal{F}_a \equiv \{f_j\}$	Set of inserted flows. J is the total number of flows. $j \in [1, \dots, J]$
$\mathcal{F}_d \equiv \{f_j\}$	Set of de-inserted flows.
$\mathcal{D}_j^{kl} \equiv \{d_j^{kl}\}$	Set of demand vectors of a flow j . This set composes of possible path through which flow may be routed. K is the total number of paths. $k \in [1, \dots, K]$. L is the total number of dimensions. $l \in [1, \dots, L]$
C_i^l	Capacity of dimension l in bin i . I is the total number of bins. $i \in [1, \dots, I]$
$\mathcal{B}_{ij}^m \equiv \{b_{ij}^m\}$	Set of possible bin vectors in which flow j can be placed. The possible bin vectors depend on the periodicity of the flow. M is the total number of possible bin vectors for flow j . $m \in [1, \dots, M]$
$V_i^l(t)$	The vector of remaining capacities of bins at time t . The vector is evaluated from current state of the bins $\mathcal{B}_s$.
w_j	Size of flow j . The size of flow is calculated as $\frac{\text{payload(bits)}}{\text{Bandwidth}}$
T_j	Periodicity of flow j .
$\mathcal{R}_j$	Redundancy level of the flow j . Redundancy level determine how many copies of flow j is transmitted via redundant disjoint paths.

VI. EXPERIMENTAL RESULTS

In this section, we present the evaluation of four developed heuristics for dynamic scheduling and routing for time-aware shaper and FRER in zonal architecture. The heuristics are implemented in python and the experiments were run on 64-bit dual cores 1.7GHz Intel(R) Core i5 processor with 8GB memory. We use response time and schedulability on varying load as a performance metric with a redundancy level of one for every flow.

The minimum and maximum period in our network (Fig. 1a) are 5ms and 50ms respectively, which corresponds to 10 bins. Each bin composes of 24 dimensions (i.e., the total number of egress ports connecting ACUs, CPU, TCU and switch to each other in the topology in Fig. 1), and each dimension has a capacity of 5ms. The bandwidth of every link is set as 1 Gbps. Each path of the flow can not be composed of more than four hops. With the limitation in path length, the end to end delay of the flows are bounded.

To measure the response time, the worst-case execution time of

the heuristics are recorded on a set of synthetic benchmarks in which all of the flows require a redundant level of one. Fault-Tolerant Coefficient of variation gives worsts timing among other heuristics as it takes 1.70s to schedule and route 500 flows as shown in Fig. 2. Since it requires more computational power to evaluate the mean and standard deviation of the bins. The other heuristics are faster and gives redundant schedules within 410ms (Fig. 2).

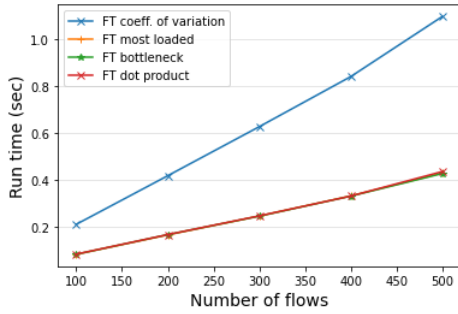


Fig. 2: Response time of Heuristics

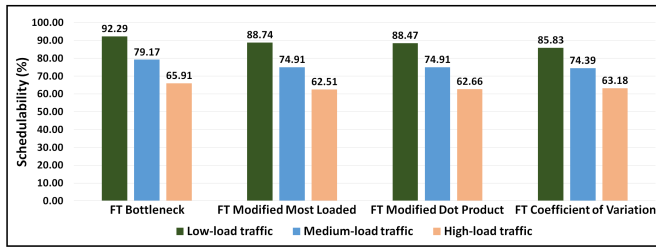


Fig. 3: Schedulability of Heuristics

We use three synthetic configurations [2], namely low-load, medium-load and high-load traffic to analyze the schedulability of developed fault-tolerant heuristics. In all of three configuration, data is sent from ACU to CPU, ACU to ACU, TCU to CPU and CPU to TCU with a periodicity of 5, 10, 25 and 50ms. The data composes of 25 KB, 31.25 KB and 37.5 KB in low, medium and high load traffic respectively. Each flow is required to schedule and route on two disjoint paths. Randomly shuffled 220 flows are given to each heuristics and average schedulable flows are measured. Interestingly, Fault-Tolerant Bottleneck Heuristic (FTBH) is ahead of other heuristics, as it schedules 3.0 – 7.0% more flows than other as shown in Fig. 3.

VII. CONCLUSION

In this paper, we have developed fault-tolerant based dynamic scheduling and routing heuristics considering FRER and time-aware shaper capabilities of IEEE TSN standard. We have transformed the problem into vector bin packing and developed four fault-tolerant based heuristics. Response time and schedulability with varying load conditions are used as a performance metric to show the efficacy of four developed heuristics. The heuristics are examined under a redundancy

level of one, such that two copies of the flow are scheduled and routed. The Fault-Tolerant Bottleneck Heuristic (FTBH) outperforms other heuristics in terms of schedulability and response time as it schedules approximately 3.0 – 7.0% more flows as compared to others. The response time shows that FTBH schedules 500 flows in approximately 410ms which is a reasonable time for dynamic reconfiguration.

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